



F1-17 Orifice and Free Jet Flow

A constant head tank is maintained with water from the Hydraulics Bench. The orifice (3mm or 6mm) is installed at the base of this tank ensuring a flush inside surface. The jet trajectory is mapped using 8-point gauges to determine the discharge coefficient.

PRODUCT CODE: F1-17

Categories: Aerospace, Educational, F1 Series Fluid Mechanics

Tags: Adjustable Probes , Armfield , Coefficient Of Velocity , F1-17 , Fluid Mechanics , Hydraulics Bench , Level Scale , Measured Trajectory , Orifice And Free Flow Jet , Trajectory Of The Jet

DESCRIPTION

The Orifice & Free Jet Flow accessory incorporates a constant head tank fed with water from the hydraulics bench. The orifice is installed at the base of this tank by means of a special wall fitting which provides a flush inside surface.

The head is maintained at a constant value by an adjustable overflow pipe and is indicated by a level scale. A series of adjustable probes enable the path followed by the jet to be ascertained.

Adjustable feet permit levelling.

TECHNICAL SPECIFICATIONS

- **Orifice diameters:** 3.0mm and 6.0mm
- **Jet trajectory probes:** 8
- **Max constant head:** 410mm

Requires Hydraulics Bench Service unit F1-10/F1-10-2

FEATURES & BENEFITS

Experimental Content

- Establishing the coefficient of velocity for a small orifice
- Finding the coefficient of discharge for a small orifice with flow under constant head and flow under varying head

- Comparing the measured trajectory of a jet with that predicted by simple theory of mechanics
- Effect of tank level on jet outlet velocity

Dimensions

Length: 0.67m

Width: 0.33m

Height: 0.60m

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